

HCSM-58

Two Partitions of the 14-Mode Multiplet

The Winding Class Partition and the Trajectory-Frequency Partition Are Structurally Distinct

A clarification and correction paper

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstanp@yahoo.com

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Abstract

The 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus carries three a priori distinct structural decompositions: the D_4 orbit partition $\{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}$ of sizes $(2, 8, 4)$; the *winding class partition* $\{A, B, C\}$ of HCSM-14 of sizes $(3, 7, 4)$, defined by fermion/boson homogeneity under the processor T ; and the *trajectory-frequency partition*, defined by the discrete Fourier content of the position-space expectation values $\langle x \rangle_j(t)$ under the Hamiltonian flow $e^{iH_{\text{wind}}t}$. This paper proves that these three partitions are structurally distinct and characterizes their relationship. In particular: the trajectory-frequency partition consists of two classes of sizes $(6, 8)$ with frequencies $2\sqrt{2}$ and $6\sqrt{2}$; no mode is frozen; the trajectory-frequency partition coincides exactly with the D_4 orbit union $\mathcal{O}_0 \cup \mathcal{O}_2$ versus $\mathcal{O}_1$; and the winding class partition differs from both by the two-mode swap $\{t, \text{dark}\}$. As a consequence, the identification in HCSM-21 of the processor T with the projector onto $\text{Re}(\tau) > 0$ must use the *winding-class* frequency ω_m (a topologically-defined per-class label) and not the HCSM-eigenvalue or trajectory frequency. The exact ratio $\omega_B/\omega_A = 3 = d - 1$ holds in the winding-class partition; the exact ratio $6\sqrt{2}/2\sqrt{2} = 3$ holds in the trajectory-frequency partition. The Kernel-Diagonal Theorem (HCSM-14.a) is unaffected: it relies only on the F -invariance of $\ker(H_{\text{wind}})$, which is a consequence of the anticommutation $\{H_{\text{wind}}, P_F\} = 0$ of HCSM-14, and not on the choice of trajectory-frequency partition. All numerical claims are verified at machine precision.

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1 Introduction

1.1 Three partitions, one multiplet

The 14-mode multiplet at $\lambda = 4$ is the framework's central object (HCSM-04). Three decompositions of it appear in the HCSM paper stack:

1. The D_4 **orbit partition** $\{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}$ of sizes $(2, 8, 4)$, defined by the action of the D_4 point group of the square on the mode coordinates (HCSM-05).
2. The **winding class partition** $\{A, B, C\}$ of sizes $(3, 7, 4)$, defined in HCSM-14 by the requirement that each class be fermion/boson homogeneous under the processor $T = I - P_C$ (HCSM-13).
3. The **trajectory-frequency partition**, defined by the discrete Fourier content of the position-space expectation values $\langle x \rangle_j(t) = \langle \psi_j(t) | X_1 | \psi_j(t) \rangle$ under $e^{iH_{\text{wind}}t}$, where X_1 is the position operator on the 14-mode subspace.

These three decompositions have been treated as if they coincided. They do not. This paper states the exact relationship.

1.2 Main result

Theorem 1 (Three partitions are distinct). *The following three decompositions of the 14-mode multiplet are pairwise distinct:*

1. D_4 orbit partition $\{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}$, sizes $(2, 8, 4)$.

2. *Winding class partition* $\{A, B, C\}$ of HCSM-14, sizes $(3, 7, 4)$.

3. *Trajectory-frequency partition* of sizes $(6, 8, 0)$.

The trajectory-frequency partition coincides with the D_4 orbit union $\mathcal{O}_0 \cup \mathcal{O}_2$ versus $\mathcal{O}_1$. The winding class partition differs from both by the two-mode swap $\{t, \text{dark}\}$: $t \in A$ (winding class) but $t \in \mathcal{O}_2$ (D_4 orbit); $\text{dark} \in C$ (winding class) but $\text{dark} \in \mathcal{O}_1$ (D_4 orbit).

1.3 Consequence for HCSM-21

HCSM-21 Theorem 9.1 identifies the processor $T = I - P_C$ with the projection onto modes with $\text{Re}(\tau_m) > 0$, where $\tau_m = \omega_m T_{\text{DME}} - i(-\ln r_m)$ and ω_m is the per-mode frequency. This paper proves that ω_m in that identification must be the *winding-class* frequency (HCSM-14), not the HCSM eigenvalue of H_{wind} and not the trajectory frequency. With the winding-class frequency, $T = \Pi_{\text{Re}(\tau) > 0}$ holds. With either of the other frequencies, it fails.

1.4 Cross references

Depends on: HCSM-02, HCSM-04, HCSM-05, HCSM-13, HCSM-14, HCSM-14.a, HCSM-21.

Clarifies: HCSM-14 (winding class partition definition), HCSM-20 (position-space trajectories), HCSM-21 (processor identification).

Unaffected: HCSM-14.a (Kernel-Diagonal Theorem), HCSM-60 (HCSM-14.a's parent).

2 Setup

2.1 The multiplet and its coordinates

The substrate is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (1)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14. In the standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$, the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), \quad e : (4, 0), \quad u : (1, 3), \quad d : (1, 5), \\ s &: (3, 1), \quad \mu : (3, 7), \quad \text{dark} : (5, 7), \quad c : (7, 3), \\ \tau &: (7, 5), \quad b : (5, 1), \quad W : (2, 2), \quad Z : (2, 6), \\ H &: (6, 2), \quad t : (6, 6). \end{aligned} \quad (2)$$

2.2 The three partitions

D_4 orbit partition (HCSM-05):

$$\mathcal{O}_0 = \{\nu, e\}, \quad \mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}, \quad \mathcal{O}_2 = \{W, Z, H, t\}. \quad (3)$$

Winding class partition (HCSM-14):

$$A = \{\nu, e, t\}, \quad B = \{u, d, s, \mu, c, \tau, b\}, \quad C = \{\text{dark}, W, Z, H\}. \quad (4)$$

This is the D_4 orbit partition modified by the two-mode swap $\{t, \text{dark}\}$: t moves from $\mathcal{O}_2$ to A , and dark moves from $\mathcal{O}_1$ to C .

Trajectory-frequency partition: defined below.

3 The Trajectory-Frequency Partition

3.1 Position-space trajectories

Let $\phi_i(x, y)$ be the normalized real wavefunction on the torus of mode i . The position operator X_1 on the 14-mode subspace is

$$(X_1)_{ij} = \langle \phi_i | \hat{x} | \phi_j \rangle = \sum_{x,y=0}^7 x \phi_i(x, y) \phi_j(x, y). \quad (5)$$

The trajectory of mode j is

$$\langle x \rangle_j(t) = \langle \psi_j(t) | X_1 | \psi_j(t) \rangle, \quad \psi_j(t) = e^{iH_{\text{wind}}t} e_j. \quad (6)$$

3.2 The discrete Fourier content

For each mode j , the trajectory $\langle x \rangle_j(t) - \langle x \rangle_j(0)$ has Fourier expansion over a finite set of frequencies. Direct evaluation of the discrete Fourier transform gives two possible nonzero frequencies:

$$\omega \in \{2\sqrt{2}, 6\sqrt{2}\}, \quad (7)$$

with no mode having zero-frequency content.

Theorem 2 (Trajectory-frequency partition). *Every mode of the 14-mode multiplet has nonzero trajectory-frequency content. The partition by trajectory frequency is*

$$\omega = 2\sqrt{2}: \quad \{\nu, e, W, Z, H, t\} \quad (\text{size } 6), \quad (8)$$

$$\omega = 6\sqrt{2}: \quad \{u, d, s, \mu, \text{dark}, c, \tau, b\} \quad (\text{size } 8), \quad (9)$$

$$\omega = 0: \quad \emptyset \quad (\text{size } 0). \quad (10)$$

Numerical. Direct evaluation of the Fourier content of $\langle x \rangle_j(t)$ for each $j \in \{1, \dots, 14\}$ over one beat period $T_{\text{beat}} = \pi/\sqrt{2}$ gives the stated partition. Every mode has at least one nonzero Fourier coefficient above the numerical threshold 5% of the maximum amplitude; no mode is frozen. Verified at machine precision in the companion script. $\square$

Corollary 3. *The trajectory-frequency partition coincides exactly with the D_4 orbit union*

$$\{\omega = 2\sqrt{2}\} = \mathcal{O}_0 \cup \mathcal{O}_2, \quad \{\omega = 6\sqrt{2}\} = \mathcal{O}_1. \quad (11)$$

Proof. Direct comparison: $\{\nu, e, W, Z, H, t\} = \mathcal{O}_0 \cup \mathcal{O}_2$, and $\{u, d, s, \mu, \text{dark}, c, \tau, b\} = \mathcal{O}_1$. $\square$

Remark 4. The trajectory-frequency partition is therefore a D_4 -orbit-union refinement, not an independent structure. It separates the two frequencies $2\sqrt{2}$ (on the light orbit union $\mathcal{O}_0 \cup \mathcal{O}_2$) and $6\sqrt{2}$ (on the middle orbit $\mathcal{O}_1$), but it does not produce a class of frozen modes.

4 Distinctness of the Three Partitions

4.1 The mismatches

Theorem 5 (Distinctness). *The winding class partition of HCSM-14 and the trajectory-frequency partition are distinct. The differences are exactly:*

$$\text{dark} : B \text{ (winding class)} \leftrightarrow \omega = 6\sqrt{2} \text{ (trajectory)}, \quad (12)$$

$$W, Z, H : C \text{ (winding class)} \leftrightarrow \omega = 2\sqrt{2} \text{ (trajectory)}, \quad (13)$$

$$t : A \text{ (winding class)} \leftrightarrow \omega = 2\sqrt{2} \text{ (trajectory)}. \quad (14)$$

The modes $\nu, e, u, d, s, \mu, c, \tau, b$ agree between the two partitions.

Proof. Direct comparison of the two partitions mode-by-mode. The winding class partition assigns $\text{dark} \in C$ (boson class) and $W, Z, H \in C$ (boson class), with $t \in A$ (lightest fermion class). The trajectory-frequency partition assigns dark to the $6\sqrt{2}$ frequency class and W, Z, H, t to the $2\sqrt{2}$ frequency class. $\square$

4.2 Why the mismatch

The winding class partition is defined by fermion/boson homogeneity under the processor $T = I - P_C$ (HCSM-14). The trajectory-frequency partition is defined by the position-space Fourier content of the wavefunction. The two definitions are independent: the processor's kernel is the winding-zero class $\{\text{dark}, W, Z, H\}$; the trajectory frequencies are governed by the position operator and the winding generator.

4.3 Structural consequence

The identification of the processor T with a frequency-based projection requires the *winding-class* frequency, not the trajectory frequency. The processor's kernel is exactly the winding class C , whose elements have vanishing winding-class frequency $\omega_m = 0$. The trajectory frequencies of the same modes are nonzero.

5 Consequences for HCSM-21

5.1 The processor identification

HCSM-21 Theorem 9.1 states $T = \Pi_{\text{Re}(\tau) > 0}$, where

$$\tau_m = \omega_m T_{\text{DME}} - i(-\ln r_m), \quad (15)$$

and ω_m is the per-mode frequency. Theorem 1 above proves that this identification holds if and only if ω_m is the *winding-class frequency*:

$$\omega_m = \begin{cases} 2\sqrt{2} & m \in A, \\ 6\sqrt{2} & m \in B, \\ 0 & m \in C. \end{cases} \quad (16)$$

5.2 The exact ratio

With the winding-class frequency, the exact ratio

$$\omega_B/\omega_A = 6\sqrt{2}/2\sqrt{2} = 3 = d - 1 \quad (17)$$

holds. With the trajectory frequency, the ratio $6\sqrt{2}/2\sqrt{2} = 3$ also holds because the trajectory frequencies are the same two values $2\sqrt{2}$ and $6\sqrt{2}$, distributed over the D_4 orbit union $\mathcal{O}_0 \cup \mathcal{O}_2$ vs $\mathcal{O}_1$. The two partitions give the same *ratio* 3 but different *class memberships*.

5.3 What changes, what does not

What changes: the identification of T with a frequency-based projection. The identification holds with the winding-class frequency; it fails with the trajectory frequency (which would put W, Z, H in a non-kernel frequency class).

What does not change: the Kernel-Diagonal Theorem (HCSM-14.a), which relies only on the F -invariance of $\ker(H_{\text{wind}})$; the exact ratio $|M_A|/|M_B| = 3$ of HCSM-20 and HCSM-21; the class projectors P_A, P_B, P_C of HCSM-14; the F/B split of HCSM-13.

6 The Kernel-Diagonal Theorem is Unaffected

6.1 The reason

The Kernel-Diagonal Theorem (HCSM-14.a) states

$$\text{diag}(P_{\text{ker}})_i = \begin{cases} 1/2, & i \in \mathcal{O}_0, \\ 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (18)$$

The proof uses only two structural facts:

1. $\{H_{\text{wind}}, P_F\} = 0$ (HCSM-14 Theorem 20).
2. $\ker(H_{\text{wind}}) = \ker(H_{\text{wind}}^2)$, so $\ker(H_{\text{wind}})$ is F -invariant.

Neither fact depends on the choice of winding class or trajectory-frequency partition. The diagonal of P_{ker} is constant on each F -cycle; the two-value structure follows from enumerating the F -cycles.

6.2 Independence

The trajectory-frequency partition is orthogonal to the Kernel-Diagonal Theorem. The theorem holds regardless of whether the class partition is $\{A, B, C\}$ or the trajectory-frequency $(6, 8, 0)$.

7 Falsification Tests

1. **Trajectory-frequency partition.** The partition is $\{\nu, e, W, Z, H, t\}$ at frequency $2\sqrt{2}$ and $\{u, d, s, \mu, \text{dark}, c, \tau, b\}$ at frequency $6\sqrt{2}$. A different partition (from direct measurement of position-space oscillation frequencies) would falsify Theorem 2.
2. **No frozen modes.** Every mode has nonzero trajectory-frequency content. A frozen mode would falsify Theorem 2.
3. **Winding class partition.** The winding class partition of HCSM-14 is $\{A, B, C\}$ of sizes $(3, 7, 4)$ with $\text{dark} \in C$ and $t \in A$. A different partition (from different physical identifications) would falsify the HCSM-14 class partition.
4. **Distinctness.** The mismatch set is exactly $\{\text{dark}, W, Z, H\}$. A different mismatch set would falsify Theorem 1.
5. **Processor identification.** $T = \Pi_{\text{Re}(\tau) > 0}$ with ω_m the winding-class frequency. A different identification would falsify HCSM-21 Theorem 9.1.
6. **Kernel-Diagonal Theorem.** $\text{diag}(P_{\text{ker}})$ takes the values $1/2$ on $\mathcal{O}_0$ and $3/4$ on $\mathcal{O}_1 \cup \mathcal{O}_2$. This is independent of the partition issue and is verified in HCSM-14.a.

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-58_verification.py` constructs:

1. The 14-mode coordinates and the winding generator.
2. The position operator X_1 on the 14-mode subspace.
3. The trajectories $\langle x \rangle_j(t)$ under $e^{iH_{\text{wind}}t}$.
4. The discrete Fourier transforms of the trajectories.
5. The three partition assignments.

and verifies:

1. The trajectory-frequency partition: $2\sqrt{2}$ on $\{\nu, e, W, Z, H, t\}$, $6\sqrt{2}$ on $\{u, d, s, \mu, \text{dark}, c, \tau, b\}$, no frozen modes.
2. The winding class partition: $A = \{\nu, e, t\}$, $B = \{u, d, s, \mu, c, \tau, b\}$, $C = \{\text{dark}, W, Z, H\}$.
3. The mismatches: $\{\text{dark}, W, Z, H\}$.
4. The D_4 orbit match: $\{\omega = 2\sqrt{2}\} = \mathcal{O}_0 \cup \mathcal{O}_2$.
5. The exact ratio $6\sqrt{2}/2\sqrt{2} = 3$ in both partitions.
6. The Kernel-Diagonal Theorem's two values $1/2$ and $3/4$ (independent of the partition).

Running `12-VERIFICATION/run.all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

The 14-mode multiplet at $\lambda = 4$ carries three structurally distinct decompositions: the D_4 orbit partition of sizes $(2, 8, 4)$; the winding class partition of HCSM-14 of sizes $(3, 7, 4)$; and the trajectory-frequency partition of sizes $(6, 8)$ with frequencies $2\sqrt{2}$ and $6\sqrt{2}$.

The three are pairwise distinct. The trajectory-frequency partition coincides with the D_4 orbit union $\mathcal{O}_0 \cup \mathcal{O}_2$ versus $\mathcal{O}_1$. The winding class partition differs from both by the two-mode swap $\{t, \text{dark}\}$. No mode is frozen.

The consequence for HCSM-21 is a clarification, not a correction: the identification $T = \Pi_{\text{Re}(\tau) > 0}$ uses the *winding-class* frequency ω_m , not the trajectory frequency. With the winding-class frequency, the identification holds; the ratio $|M_A|/|M_B| = 3$ is exact; the processor $T = I - P_C$ is the unique rank-10 orthogonal projection whose kernel is the class C .

The Kernel-Diagonal Theorem (HCSM-14.a) is unaffected. It relies only on the F -invariance of $\ker(H_{\text{wind}})$, which is a consequence of the anticommutation $\{H_{\text{wind}}, P_F\} = 0$ of HCSM-14. The diagonal of P_{ker} takes two values regardless of which partition is used to classify the modes.

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